

MA 301  
Test 1  
Spring - 2015

| Problem | points | out of |
|---------|--------|--------|
| 1       |        | 8      |
| 2       |        | 6      |
| 3       |        | 6      |
| 4       |        | 3      |
| 5       |        | 3      |
| 6       |        | 2      |
| 7       |        | 2      |
| Score   |        | 30     |

Name: \_\_\_\_\_

HR: \_\_\_\_\_

Signature: \_\_\_\_\_

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| <b>READ BEFORE YOU START</b>                                                                                                                                                                                                                                                                                                                                                                             |
| <ul style="list-style-type: none"><li>* Show all work.</li><li>* Change your calculator mode to RADIANS.</li><li>* Sloppy solutions may be penalized.</li><li>* Use calculators ONLY for computations.</li><li>* You must write up every problem in a clear and organized way.</li><li>* If you write solutions on a page different than the question, then clearly state where it is located.</li></ul> |

**Problem 1:** Given  $\mathbf{F} = [2, 2y + z, y + 1]$

(a) Show that  $\mathbf{F}$  is conservative in two different ways

(b) Assuming  $\mathbf{F}$  is a force, compute the work required to move a mass from  $(-1, 4, 2)$  to  $(0, 0, -6)$

**Problem 2:** Suppose  $\mathbf{F}(x, y) = [xy - x^{15}, 2x + \ln y]$  and suppose  $C$  consists of the following 3 curves: (i) the line segment from  $(0, 0)$  to  $(2, 0)$ , (ii) the curve  $y = 4 - x^2$  from  $(2, 0)$  to  $(0, 4)$  and (iii) the line segment from  $(0, 4)$  to  $(0, 0)$ .

(a) Sketch  $C$

(b) Compute  $\int_C \mathbf{F} \cdot d\mathbf{r}$

**Problem 3:** Let  $\mathbf{F} = [z^2, x - y, 7]$  and  $C$  be the intersection of the surfaces  $x = y + z$  and  $y = z^2$  with  $0 \leq y \leq 9$ .

Compute  $\int_C \mathbf{F} \cdot d\mathbf{r}$

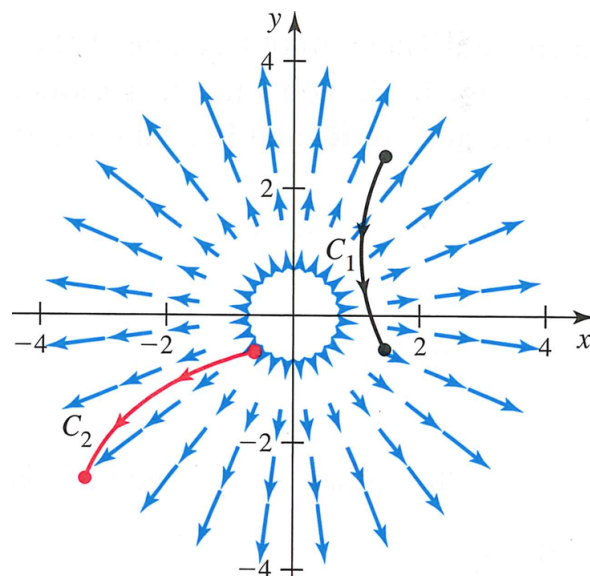
**Problem 4:** Answer the following true/false questions

- (a) True or False: If  $C$  is an ellipse and  $\mathbf{F}$  is irrotational then  $\int_C \mathbf{F} \cdot d\mathbf{r} = 0$
- (b) True or False: If  $\mathbf{F} = \nabla f$  then  $\mathbf{F}$  is conservative.
- (c) True or False: Line integrals over all constant vector fields are path-independent.

**Problem 5:** Let  $\mathbf{F}$  be the vector field and  $C_1$  and  $C_2$  be the curves shown in the provided figure.

- (a) Would the work for a mass traveling along  $C_1$  be positive, negative, or zero? Explain.

- (b) Would the work for a mass traveling along  $C_2$  be positive, negative, or zero? Explain.



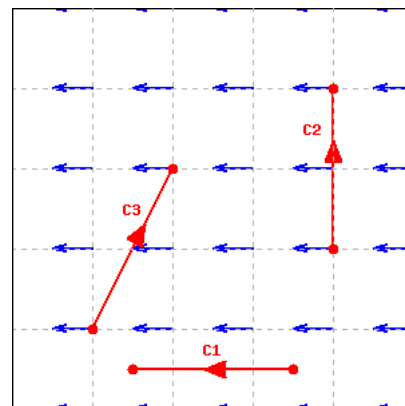
- (c) Which is larger:  $\|\text{curl}(\mathbf{F}(2,2))\|$  or  $\text{div}(\mathbf{F}(2,2))$ ? Explain.

**Problem 6:** Given the figure, answer the following:

- (a) List the line integrals  $\int_{C_1} \mathbf{F} \cdot d\mathbf{r}$ ,  $\int_{C_2} \mathbf{F} \cdot d\mathbf{r}$ ,  $\int_{C_3} \mathbf{F} \cdot d\mathbf{r}$  in increasing order value.

- (b) Circle the integrals that are path independent

$$\int_{C_1} \mathbf{F} \cdot d\mathbf{r} \quad \int_{C_2} \mathbf{F} \cdot d\mathbf{r} \quad \int_{C_3} \mathbf{F} \cdot d\mathbf{r}$$



**Problem 7:** Let  $\mathbf{F} = \nabla(\ln x + (x - y)^2)$  and  $C$  be parameterized by  $\mathbf{r}(t) = [e^t, t^2 - t]$  where  $0 \leq t \leq 2$ .

Compute  $\int_C \mathbf{F} \cdot d\mathbf{r}$